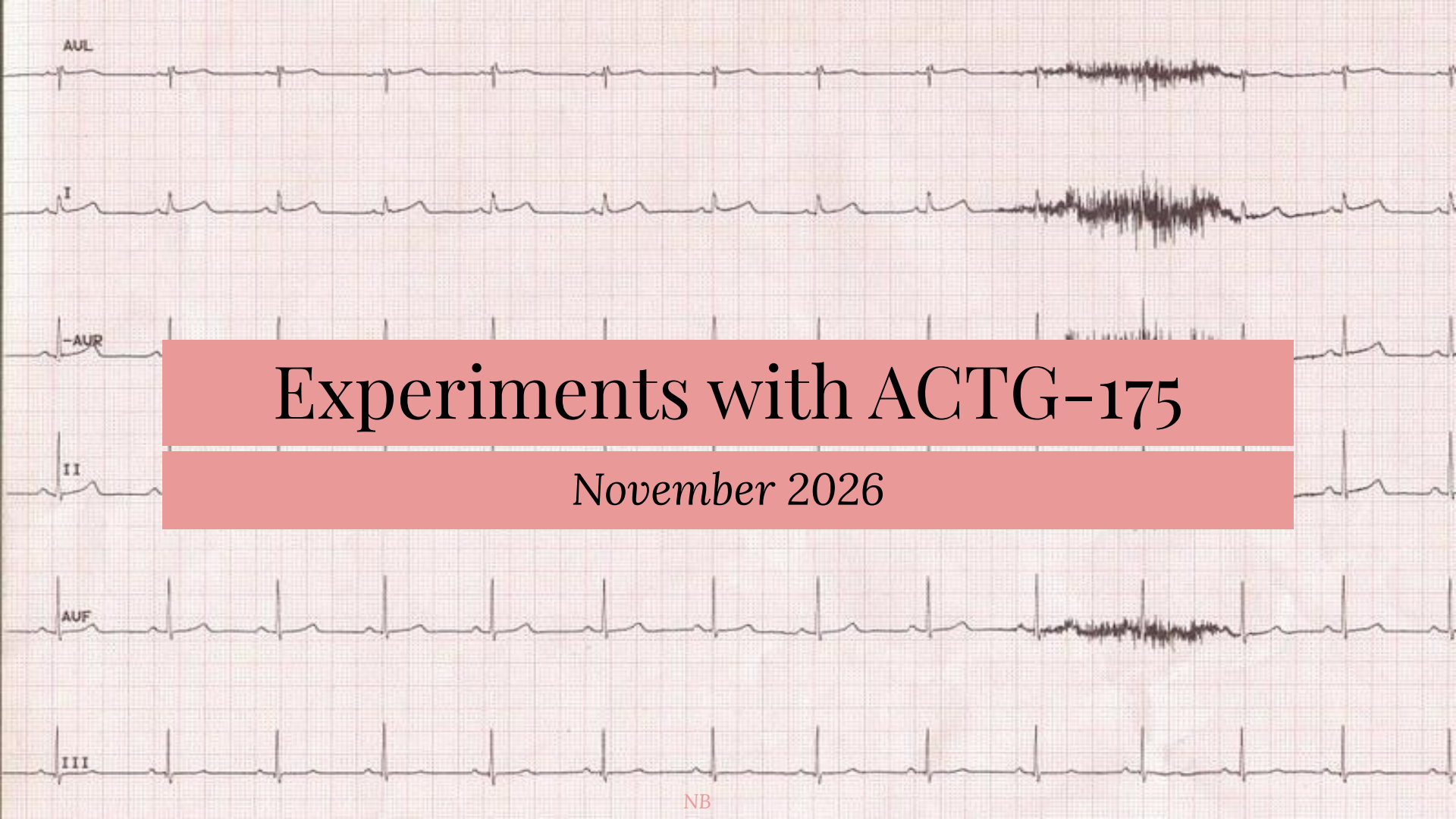


Experiments with ACTG-175

November 2026

Datasets

- The **AIDS Clinical Trial Group 175 (ACTG-175)** dataset is publically available
- The *treat* variable shows whether a patient underwent **zidovudine (ZDV)** monotherapy or not.
  The *arms* variable further breaks it down into ZDV monotherapy and the three increasing doses of **Didanosine (ddl)**

Arm #	0	1	2	3
Arm	**ZDV 600**	**ZDV (600mg) + dd (400mg)**	**ZDV (600mg) + ZAL (2.25mg)**	**dd (400mg)**
Patient Count	532	522	524	561

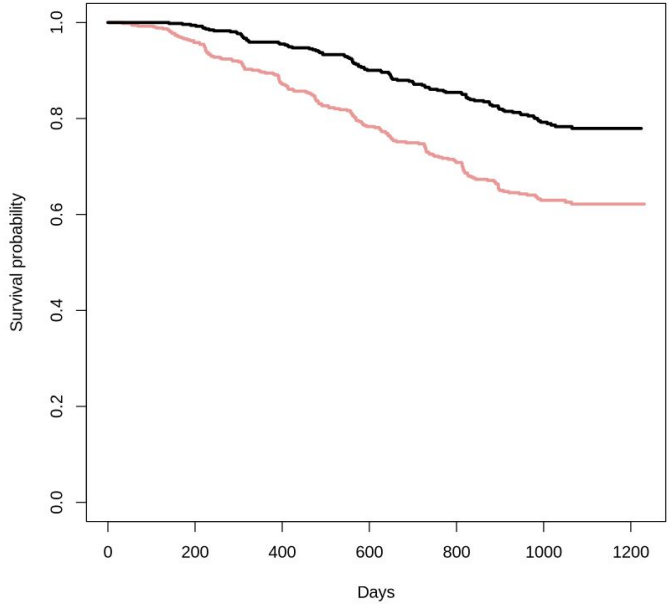
Dataset of interest

NB

Survival analysis

Survival analysis (Treated v. untreated)

- Here, **treated *means* additional dd 200mg on top of ZDV 600mg**
- Dataset of 1054 patients
- Some variables of note:
  - Treated or untreated
  - Weight at baseline
  - Gender
  - Age
  - Karnof scale - ability to tolerate treatments
  - History of IV drug use
- [*Pictured:*] Kaplan Meier curve showing different risks for treated and untreated patients



NB

Survival analysis (Cox)

- Cox PH test : p-value of 0.166 when fit on the above
  i.e. satisfied the Proportional Hazards assumption

- Of the selected variables, treatment had a significant impact, so did response to treatment (karnof as proxy)

- i.e.
  "Both treatment and ability to tolerate harsher treatments reduced risk of death
  With treatment showing a 50% reduced hazard"

Variable	HR (95% CI)	p.value
treat	0.50 (0.39–0.64)	<0.001
wtkg	1.01 (1.00–1.02)	0.1686
gender	1.34 (0.94–1.92)	0.1046
age	1.00 (0.99–1.02)	0.5379
karnof	0.98 (0.96–1.00)	0.0229
drugs	0.68 (0.45–1.04)	0.0728

NB

Survival analysis (Cox) - interaction

- An obvious follow up was to ask if there was an interaction between treatment and ability to tolerate such treatments
- There was no additional impact of being a tolerant treated patient
- So, is there some high yield / high responder group that got treated?
  Causal forests are useful here

Variable	HR (95% CI)	p.value
treat	4.62 (0.11–200.96)	0.4267
karnof	0.99 (0.96–1.01)	0.2508
wtkg	1.01 (1.00–1.02)	0.1725
gender	1.36 (0.95–1.95)	0.0882
age	1.00 (0.99–1.02)	0.5307
drugs	0.68 (0.45–1.04)	0.0761
treat:karnof	0.98 (0.94–1.02)	0.2483

Causal Forests

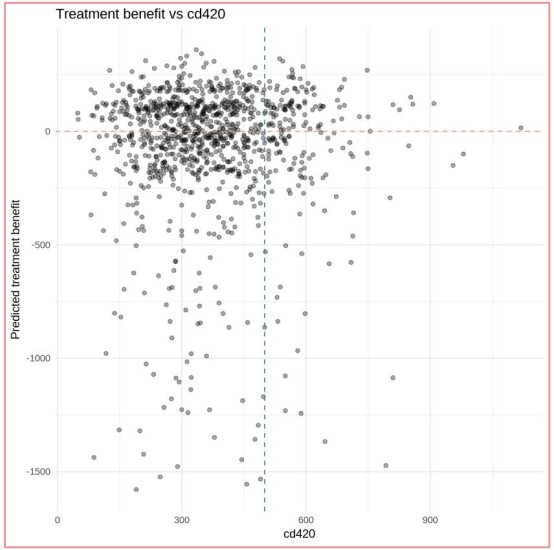
Why a causal forest?

- A causal forest estimates a value called CATE (Conditional Average Treatment Effect)
- Average because it takes patients within a group (say, white 80 year-old men with diabetes, and compares the mean of the treated and untreated members who belong to this group)
- Clinical trials are especially well suited to this, since the patients are randomized for much this reason.
- Since we're explicitly working with survival data, we'll be using the causal *survival* forest specifically.

What does the high-benefit group look like?

When the model was fit on just **'IV drug use history', 'karnof score', and 'weight in kg'**

	Treated	**Untreated**
Zero/ Neg. Benefit	267	255
Positive Benefit	265	267

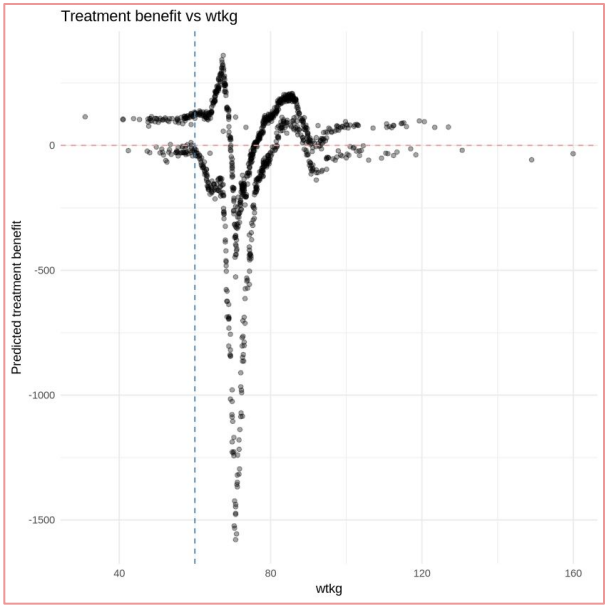


NB

Which variables mattered most in that fit?

When the model was fit on just **'IV drug use history', 'karnof score', and 'weight in kg'**

Variable	**Importance**
Wt. kg	0.734
Karnof	0.264
Drug use	0.001



NB

Now, with carefully chosen variables: